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Week 3 Intro

Lesson 1: Analysis of the Sample Mean

Lesson 2: General Framework of Statistical Estimation

Lesson 3: Central Limit Theorem

Video: The Central Limit Theorem I 5 min

Video: The Central Limit Theorem II 5 min

Graded Assignment: Learning Check - The Central Limit Theorem Submitted

Weekly Assignment

Your grade: 100%

Your latest: 100% > Your highest: 100%

Next item →

Learning Check - The Central Limit Theorem

Instructions

Review Learning Objectives

1 / 1 point

1. Central Limit Theorem (CLT) applies to small sample sizes.

☐ True

COACH

☒ False

Ready to review what you've learned before starting the assignment? I'm here to help.

☒ Correct

Help me practice

Let's chat

CLT applies only to sufficiently large sample sizes.

Assignment details

2. Let $X_1, X_2, \dots, X_n$ be a sequence of iid random variables and $\bar{X}$ be the sample mean. We know that $\bar{X}$ is also a Gaussian. What is its distribution? 1 / 1 point

☐ $N(\frac{\mu}{n}, \sigma^2)$

Submitted

Sep 27, 6:03 PM IST

☒ $N(\mu, \sigma^2)$

COACH

☐ $N(\frac{\mu}{n}, \frac{1}{n})$

To pass you need at least 80%. We keep your highest score.

☒ Correct

Review "The Central Limit Theorem I" for full explanation.

Like

Dislike

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See feedback

3. Let $\Phi_{\mu, \frac{1}{n}}$ denote the CDF of $N(\mu, \sigma^2)$ and let $F_{\bar{X}}(x)$ denote the CDF of $\bar{X}$. Which of the following is true? 1 / 1 point

☐ $F_{\bar{X}}(x) - \Phi_{\mu, \frac{1}{n}} \leq \frac{C}{\sqrt{n}}$

☒ $|F_{\bar{X}}(x) - \Phi_{\mu, \frac{1}{n}}| \leq \frac{C}{\sqrt{n}}$

☐ $|\Phi_{\mu, \frac{1}{n}} - F_{\bar{X}}(x)| \leq \frac{C}{\sqrt{n}}$

☐ $\Phi_{\mu, \frac{1}{n}} - F_{\bar{X}}(x) \leq \frac{C}{\sqrt{n}}$

☒ Correct

See "The Central Limit Theorem II" for a full explanation.

https://www.coursera.org/learn/statistics-for-data-science-essentials/assignment-submission/3xogt/learning-check-the-central-limit-theorem/view-feedback

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